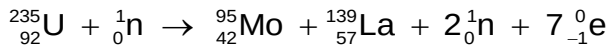


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Data for a nucleus and some particles are given in the table.

nucleus or particle	mass / u
$^{139}_{57}\text{La}$ nucleus	138.955
^1_0n particle	1.00863
^1_1p particle	1.00728
$^0_{-1}\text{e}$ particle	5.49×10^{-4}

What is the binding energy per nucleon of lanthanum-139 ($^{139}_{57}\text{La}$)?

A	7.84 Mev	B	7.87 MeV	C	19.13 MeV	D	19.19 MeV
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Answer: A

Given the nuclear mass and not atomic mass,

$$\Delta m = (82 \times 1.00863u) + (57 \times 1.00728u) - 138.955u = 1.16762u$$

$$\text{Energy} = \Delta mc^2 = (1.16762 \times 1.66 \times 10^{-27}) \times (3.00 \times 10^8)^2$$

$$= 1.74442428 \times 10^{-10} \text{ J} = \frac{1.74442428 \times 10^{-10}}{1.6 \times 10^{-13}} \text{ MeV} = 1090.265 \text{ MeV}$$

$$\text{Energy per nucleon} = \frac{1090.265}{139} = 7.84 \text{ MeV}$$

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The half-life of a certain radioactive isotope is 32 hours.